XINGZHE HE

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Homepage

EDUCATION

University of British Columbia, Canada *2020-2024*
Ph.D. in Computer Science
advised by Professor Helge Rhodin

Rutgers University, USA *2017-2019*
Master of Science in Data Science

University of Liverpool, UK *2015-2017*
Bachelor of Science with Honors in Mathematics with Finance

Xi'an Jiaotong-Liverpool University *2013-2015*
Bachelor of Economics with Honors in Financial Mathematics

EXPERIENCE

Descript
Research Tech Lead *Nov 2025 -*

- Lead video and audio generation research, and video editing agent.

Applied Research Scientist *Mar 2024 - Nov 2025*

- Project leader and main contributor of video translation and video overdub models.
- Project leader of audio-video evaluation metric models.
- Model designer of audio imputation models.

Adobe
Machine Learning Intern, advised by Zhiwen Cao and Ratheesh Kalarot *Jun 2023 - Sep 2023*

- Developing generative models, particularly diffusion personalization models.

Flawless AI
Engineering Research Intern, advised by Pablo Garrido and Gaurav Bharaj *Jun 2022 - Nov 2022*

- Detect 3D keypoint from single static images with few-shot 2D keypoint annotations.
- Model mouth area with sparse 3D keypoints.

Dartmouth College
Visiting Researcher, advised by Professor Bo Zhu *Jan 2019 - Aug 2020*

- Used statistics for computer graphics and animation.
- Gave tutorials on computer vision to visiting students and undergrad students.

SELECTED PUBLICATIONS

1. A Data Perspective on Enhanced Identity Preservation for Diffusion Personalization [pdf]
   Xingzhe He, Zhiwen Cao, Nicholas Kolkin, Lantao Yu, Kun Wan, Helge Rhodin, Ratheesh Kalarot
   WACV 2025
2. Unsupervised keypoints from pretrained diffusion models [pdf] [project]
   Eric Hedlin, Gopal Sharma, Shweta Mahajan, **Xingzhe He**, Hossam Isack, Abhishek Kar Helge Rhodin, Andrea Tagliasacchi, Kwang Moo Yi
   CVPR 2024
3. Few-shot Geometry-Aware Keypoint Localization [pdf] [project]
   Xingzhe He, Gaurav Bharaj, David Ferman, Helge Rhodin, Pablo Garrido
   CVPR 2023
4. LatentKeypointGAN: Controlling GANs via Latent Keypoints [pdf] [project]
   Xingzhe He, Bastian Wandt, Helge Rhodin
   CRV 2023 (Best Paper Award)
5. AutoLink: Self-supervised Learning of Human Skeletons and Object Outlines by Linking Keypoints [pdf] [project]
   Xingzhe He, Bastian Wandt, Helge Rhodin
   NeurIPS 2022 (Spotlight $\sim 3\%$ acceptance rate)
6. GANSeg: Learning to Segment by Unsupervised Hierarchical Image Generation [pdf]
   Xingzhe He, Bastian Wandt, Helge Rhodin
   CVPR 2022
7. Nonseparable Symplectic Neural Networks [pdf] [project]
   Shiying Xiong, Yunjin Tong, **Xingzhe He**, Shuqi Yang, Cheng Yang, Bo Zhu
   ICLR 2021
8. Symplectic Neural Networks in Taylor Series Form for Hamiltonian Systems [pdf] [project]
   Yunjin Tong*, Shiying Xiong*, **Xingzhe He**, Guanghan Pan, Bo Zhu
   Journal of Computational Physics
9. Learning Physical Constraints with Neural Projections [pdf] [project]
   Shuqi Yang, **Xingzhe He**, Bo Zhu
   NeurIPS 2020
10. AdvectiveNet: An Eulerian-Lagrangian Fluidic Reservoir for Point Cloud Processing [pdf]
    Xingzhe He, Helen L. Cao, Bo Zhu
    ICLR 2020
11. Soft Multicopter Control using Neural Dynamics Identification [pdf] [video]
    Yitong Deng, Yaorui Zhang, **Xingzhe He**, Shuqi Yang, Yunjin Tong, Michael Zhang, Daniel M. DiPietro, Bo Zhu
    CoRL 2020
12. Neural vortex method: from finite lagrangian particles to infinite dimensional eulerian dynamics [pdf]
    Shiying Xiong, **Xingzhe He**, Yunjin Tong, Yitong Deng, Bo Zhu
    Computers & Fluids